Conflict-free solution sets for en-route air traffic control

Exact closest-point-of-approach matrices, control-space level sets and Hamilton–Jacobi reachability in a controller training simulator

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October 2026

Abstract

We present the conflict solver of an en-route air traffic control training simulator. Given the predicted four-dimensional trajectories of every aircraft in a sector, the solver returns, for each predicted loss of separation, the *set* of clearances — speed changes, direct routings, delaying vectors with a resume, level changes, or pairs of them — that leave the aircraft concerned separated from *every* other aircraft, not only from the one it conflicts with, ranked in the order a controller would use them. Trajectories are modelled as piecewise-linear functions of time; on this class the closest point of approach of two aircraft, restricted to the times at which they are vertically too close, has an exact closed form, computed leg by leg in time linear in the number of breakpoints. Collecting it over all pairs gives conflict matrices D^* and T^* of the situation. A clearance is a point u in a control space, and the solution set of an aircraft is the super-level set $\{u : \Phi(u) \geq S_h + m\}$ of the function $\Phi(u)$, the closest any other aircraft comes to its trajectory under u. Close in, where turn dynamics matter, a Hamilton–Jacobi reachability table tells whether any turn can still separate a pair in time. The best solutions are confirmed by running a copy of the full simulation ahead. Traffic rules of the unit are part of the model: the aircraft maintaining its level, or the first to enter the sector, keeps priority in it, solutions are taken speed first, then heading, then level, and a level the pilot did not request is asked first. We give the formulation, the algorithms, their complexity and numerical validation against sampling and simulation, and evaluate the solver as an autonomous controller on recorded training scenarios.

1 Introduction

An en-route controller separates aircraft by issuing clearances: a level, a heading, a direct routing to a point further along the route, or a speed. For each pair of aircraft that will come closer than the separation minima the controller chooses which aircraft to move and how, under conventions that are as much a part of the job as the geometry: an aircraft established at its level is not moved out of it if another solution exists; a speed adjustment is preferred to a vector, a vector to a level change; a level the pilot did not ask for is offered, not imposed; and a solution must not create a new conflict with a third aircraft.

The simulator this work is part of trains controllers of the Agadir Area Control Centre. It flies every aircraft with type performance, wind, lateral navigation and simulated pilots who read back clearances or refuse those they cannot fly. It offers a solo mode, in which a trainee controls a sector alone, and an *AI lab*, in which a language model controls a sector headlessly and its decisions are evaluated. Both need the same thing: a reference answer to "what are the good solutions here?" — to show the trainee, to guide the model, and to grade the decisions of either.

Conflict detection and resolution has a long literature, surveyed by Kuchar and Yang [1]: geometric methods based on the closest point of approach and velocity obstacles [2, 3], optimisation over the manoeuvres of several aircraft [4, 5], and reachability analysis by Hamilton–Jacobi equations [6, 7], solved with level-set methods [8]. We do not claim a new algorithm in each part; the contribution is a coherent, fast and verifiable pipeline whose output is a *solution set* rather than a single manoeuvre, built on the same trajectory prediction the radar uses, constrained by operational conventions, and checked against the simulation itself:

- an exact closest-point-of-approach (CPA) computation for piecewise-linear 4D trajectories with vertical windows and an RVSM-dependent vertical minimum, and the conflict matrices of a situation (Section 4);
- velocity obstacles over a heading grid as a fast picture of the free headings (Section 5);
- the solution set as a super-level set of a separation function over a discrete control space, with a cost encoding the controller's order of preference and the unit's priority and asking rules, and pair combinations when no single clearance works (Section 6);
- a Hamilton–Jacobi reachability table for close-range pairs, telling whether any turn can still separate them (Section 7);
- confirmation by simulation, integration in the solo radar and the AI lab, and an evaluation (Sections 8 to 10).

A design point deserves emphasis. A pairwise method — including a pairwise Hamilton–Jacobi value function — answers "does this manoeuvre separate i from j?" but not "does it create a conflict with k?". The solution set here is defined against *all* predicted trajectories, so a clearance that solves one conflict by creating another is excluded by construction; the reachability table is used only as a filter on the pair at close range, where the straight-line prediction is least reliable.

2 Setting and notation

Plane and aircraft. Positions are expressed in a conformal projection of the area in nautical miles, x east and y north. Aircraft i has position $\boldsymbol{p}_i \in \mathbb{R}^2$, altitude z_i (ft), track ψ_i (degrees clockwise from north), ground speed g_i (kt, true airspeed composed with the wind), true airspeed V_i, vertical speed $\dot{z}_i$, its route ahead $R_i = (w_1, w_2, \dots)$ (named points), and the clearances entered in the system (cleared level, heading, direct, speed). The unit vector of a track ψ is $\boldsymbol{e}(\psi) = (\sin\psi, \cos\psi)$.

Separation minima. The horizontal minimum S_h is a property of the sector (5 or 7 NM in the scenarios used here). The vertical minimum depends on the altitudes:

$$S_v(z_i, z_j) = \begin{cases} 2000 \text{ ft} & \text{if } \min(z_i, z_j) \geq \text{FL290 and } \big(\max(z_i, z_j) > \text{FL410} \\ & \qquad \text{or one aircraft is non-RVSM}\big), \\ 1000 \text{ ft} & \text{otherwise.} \end{cases} \tag{1}$$

Two aircraft are in *loss of separation* at time t when $\|\boldsymbol{p}_i(t) - \boldsymbol{p}_j(t)\| < S_h$ and $|z_i(t) - z_j(t)| < S_v$. A *predicted conflict* is a predicted loss within a horizon H ($H = 600$ s in the solver).

Operational conventions. The unit's conventions, which the solver encodes (Section 6), are:

(P) *Priority*: of two aircraft in conflict, one maintaining its level has priority in it over one climbing or descending; between two level aircraft, the one that entered the sector first keeps its level.

(O) *Order*: speed adjustment, then a heading (a delaying vector, then resume) or a direct, then a level.

(A) *Asking*: a level the aircraft did not request is offered first ("able FL370?") and given only after an affirmative — except the level it requested, its flight-plan cruise level, a descent of an arrival, or an urgent conflict (under 2 minutes) or a loss in progress.

(N) *No new conflict*: a solution must keep the moved aircraft separated from everyone.

3 Trajectory model

Definition 1 (Piecewise-linear 4D trajectory)**.** *A trajectory over* $[0, H]$ *is a finite increasing sequence of breakpoints* $0 = t_0 < t_1 < \cdots < t_K = H$ *with positions* $(\boldsymbol{p}_k, z_k)$*; between breakpoints, position and altitude are affine in time.*

This is the prediction the radar's medium-term conflict detection makes, written in a form that admits exact geometry. It is built from three pieces, merged on the union of their breakpoints.

Horizontal path. The route ahead (or the clearance) gives a polyline $P_0 = \boldsymbol{p}_i, P_1, \ldots, P_m$, extended beyond its last point along its last bearing. The aircraft walks it at ground speed $g_0 = g_i$ until the time t_s of a speed change and at g_1 after it; breakpoints are placed at the vertices, at t_s and at H.

Clearances as operators. A clearance u maps the predicted trajectory X_i to X_i^u:

- *Speed* to a Mach number or an indicated airspeed: with f the ratio of the new to the present true airspeed at the aircraft's level and temperature, the wind is unchanged, so the ground speed changes by the same amount as the true airspeed:

$$g_1 = \max\big(100,\ g_0 + V_i\,(f-1)\big), \qquad t_s = \delta_s = 40\ \text{s}, \tag{2}$$

  the time by which most of the change is achieved.

- *Heading* ψ' for a duration τ, then resume: with the turn $\Delta\psi = |\psi' - \psi_i|$ (wrapped to $[0, 180]$), a pilot delay $\delta_p = 6\,\text{s}$ and a standard-rate turn $\omega = 3^\circ/\text{s}$, the turn is modelled as half on the old and half on the new track:

$$t_\psi = \delta_p + \frac{\Delta\psi}{2\omega}, \quad P_1 = \boldsymbol{p}_i + \tfrac{g_0}{3600}\,t_\psi\,\boldsymbol{e}(\psi_i), \quad P_2 = P_1 + \tfrac{g_0}{3600}\,\tau\,\boldsymbol{e}(\psi'), \tag{3}$$

  then direct to the route point after the next one and the route on. Without τ (an open heading) the aircraft continues on ψ'.

- *Direct* to a point w_k of the route: the polyline $\boldsymbol{p}_i, w_k, w_{k+1}, \ldots$.

- *Level* L: the vertical profile below.

Turns in progress. When the aircraft's track differs from the bearing of the first leg — it has just passed a point, or a direct was given — the same half-turn construction is applied: the polyline starts with a segment of length $\frac{g_0}{3600}\,\Delta\psi/(2\omega)$ on the present track. Without it the prediction cuts the corner, by about half a mile for a 40° turn at 450 kt.

Vertical profile. With $\Delta z = 100L - z_i$, the rate r is the present vertical speed when it is in the right direction, above 300 ft/min, and no new clearance is being given; otherwise the aircraft's own rate from its performance model: the climb rate available at its level, mass and speed schedule, or its idle descent rate (times the speedbrake factor 1.5 when it is high on its profile). With a pilot delay δ for a new level,

$$z(t) = \begin{cases} z_i & t \le \delta \\ z_i + \operatorname{sgn}(\Delta z)\, r\,(t-\delta)/60 & \delta < t < t_L \\ 100L & t \ge t_L \end{cases} \qquad t_L = \delta + 60\,|\Delta z|/r. \tag{4}$$

Since the rate the aircraft will actually fly is uncertain, a candidate level change is checked at two bounds as well: a *fast* rate, $\max(r, r_{\text{nom}})$ and, in descent, with speedbrakes; and a *slow* rate, $\min(r, r_{\text{nom}})$, with the nominal $r_{\text{nom}} = 1500$ ft/min in climb and 2000 in descent. This is exact for every rate in between:

Proposition 1 (Rate bounds)**.** *Let the aircraft cross the level band of another aircraft, entering it after a height h_a and leaving it after $h_b > h_a$ of climb or descent, at a rate $r \in [r_s, r_f]$. If $h_b/r_f \ge h_a/r_s$ (the windows of the two bounds overlap), the smallest horizontal distance while vertically too close, over all rates in the interval, is the smaller of its values at r_s and at r_f.*

Proof. The horizontal paths do not depend on r. At rate r the window is $[h_a/r,\ h_b/r]$ (after the delay), so the union over $r \in [r_s, r_f]$ is $[h_a/r_f,\ h_b/r_s]$, which equals $[h_a/r_f, h_b/r_f] \cup [h_a/r_s, h_b/r_s]$ when the two overlap. The minimum of the horizontal distance over a union of windows is the smaller of the minima. □

For $S_v = 1000$ ft the band is 2000 ft deep and the condition is $r_f/r_s \le 1 + 2000/h_a$: with the widest bounds used here ($r_f/r_s \approx 1.8$) it holds for a band entered within about 2400 ft of the start of the climb or descent — the crossings that matter most, being the soonest. Beyond, the check at the two bounds can miss an intermediate rate; more rates would close the gap.

Remark 1. *The model is deliberately the one the controller's tools use: what is checked is the conflict the system predicts. Its errors — turn geometry, acceleration, climb performance, wind changes along the path — are absorbed by a margin m on the horizontal minimum and, for the best solutions, removed by confirmation in the full simulation (Section 8).*

4 Exact closest point of approach

Let X_i, X_j be piecewise-linear trajectories over $[0, H]$ and $U = \{u_0 < \cdots < u_N\}$ the union of their breakpoints. On each interval $[u_k, u_{k+1}]$, of length L_k, both are affine, hence so is their difference:

$$\boldsymbol{r}(u_k + s) = \boldsymbol{r}_k + \boldsymbol{v}_k s, \qquad \Delta z(u_k + s) = \Delta z_k + w_k s, \qquad s \in [0, L_k], \tag{5}$$

with $\boldsymbol{r} = \boldsymbol{p}_j - \boldsymbol{p}_i$, $\Delta z = z_j - z_i$.

Vertical window. The aircraft are vertically too close on the interval for s in

$$W_k = \{s \in [0, L_k] : |\Delta z_k + w_k s| < S_v\} = \begin{cases} [0, L_k] \text{ if } |\Delta z_k| < S_v, \text{ else } \emptyset & w_k = 0, \\ \big[\max(0, s_-),\ \min(L_k, s_+)\big] & w_k \neq 0, \end{cases} \tag{6}$$

where $s_\mp$ are the sorted roots $(\pm S_v - \Delta z_k)/w_k$. Since S_v depends on the altitudes through (1), the solver takes on each interval the larger of its values at the two ends, which is conservative.

Lemma 1 (CPA on a window)**.** *On a nonempty window $W_k = [a, b]$, the squared distance $\|\boldsymbol{r}_k + \boldsymbol{v}_k s\|^2$ is a convex quadratic in s, so*

$$d_k = \min_{s \in W_k} \|\boldsymbol{r}_k + \boldsymbol{v}_k s\| = \big\|\boldsymbol{r}_k + \boldsymbol{v}_k\, s_k^*\big\|, \qquad s_k^* = \mathrm{clamp}\Big(-\frac{\boldsymbol{r}_k \cdot \boldsymbol{v}_k}{\|\boldsymbol{v}_k\|^2},\, a,\, b\Big) \tag{7}$$

($s_k^* = a$ when $\boldsymbol{v}_k = 0$). The first time in W_k at which the distance falls below S_h is a if $\|\boldsymbol{r}_k + \boldsymbol{v}_k a\| < S_h$, otherwise the smaller root of $\|\boldsymbol{v}_k\|^2 s^2 + 2(\boldsymbol{r}_k \cdot \boldsymbol{v}_k)s + \|\boldsymbol{r}_k\|^2 - S_h^2 = 0$ when it lies in W_k.

Definition 2 (Pair CPA)**.** *$d_{ij}^* = \min_k d_k$ (with $d_k = +\infty$ on an empty window), attained at t_{ij}^*; the pair is in conflict iff $d_{ij}^* < S_h$; t_{ij}^f is the first infringement time. The geometry alone (all windows full) gives d_{ij}^h, the closest horizontal approach at any level.*

Proposition 2 (Exactness and cost)**.** *For piecewise-linear trajectories, d_{ij}^* and t_{ij}^f are exact (up to the conservative choice of S_v per interval), and their computation takes $O(K_i + K_j)$ operations.*

Proof. The minimum over $[0, H]$ restricted to vertical proximity is the minimum over k of the minima over W_k; each is given by Lemma 1. The merge of two sorted breakpoint lists and one constant-time step per interval give the bound. □

Sampling, by contrast. A detector that evaluates distances every Δ seconds misses part of the geometry. On an interval where the window contains the sampling cell around s^*, the sampled minimum d_Δ satisfies

$$d^* \le d_\Delta \le \sqrt{d^{*2} + \big(\|\boldsymbol{v}\|\Delta/2\big)^2} \le d^* + \|\boldsymbol{v}\|\,\Delta/2, \tag{8}$$

because the nearest sample is at most $\Delta/2$ from s^* and $\boldsymbol{r}_k + \boldsymbol{v}_k s^* \perp \boldsymbol{v}_k$ at an interior minimum. Two aircraft closing at 900 kt sampled every 20 s can thus be seen up to 2.5 NM further apart than they will be, and a vertical window shorter than Δ — an aircraft climbing through another's level — can be missed altogether. Section 10 measures both effects.

Conflict matrices. Over the n aircraft of a situation, the matrices $D^* = (d_{ij}^*)$, $T^* = (t_{ij}^*)$ and $C = (d_{ij}^* < S_h)$ summarise it: the conflicts are the true entries of C ordered by t^f. Their cost is $O(n^2 \bar{K})$. An instantaneous variant uses the present velocities: with $\boldsymbol{r} = \boldsymbol{p}_j - \boldsymbol{p}_i$ and $\boldsymbol{v} = \dot{\boldsymbol{p}}_j - \dot{\boldsymbol{p}}_i$,

$$T_{ij}^{\mathrm{s}} = \mathrm{clamp}\Big(-\frac{\boldsymbol{r} \cdot \boldsymbol{v}}{\|\boldsymbol{v}\|^2}, 0, H\Big), \qquad D_{ij}^{\mathrm{s}} = \big\|\boldsymbol{r} + \boldsymbol{v}\, T_{ij}^{\mathrm{s}}\big\|, \tag{9}$$

vectorised over all pairs at once; it is what the velocity obstacles below are made of.

5 Velocity obstacles

For the aircraft i to move and a candidate velocity $\boldsymbol{u}(\psi, g) = g\,\boldsymbol{e}(\psi)$, the relative velocity of j is $\boldsymbol{v}_r = \dot{\boldsymbol{p}}_j - \boldsymbol{u}$ and the straight-line CPA is that of the previous section. The *velocity obstacle* of j is

$$\mathrm{VO}_j = \{(\psi, g) : \|\boldsymbol{r} + \boldsymbol{v}_r\, t^*(\psi, g)\| < S_h\}, \qquad t^* = \mathrm{clamp}\big(-\boldsymbol{r}\cdot\boldsymbol{v}_r/\|\boldsymbol{v}_r\|^2, 0, L\big), \tag{10}$$

taken over the aircraft j vertically too close to i (or either changing level). In velocity space it is the classical truncated cone of apex $\dot{\boldsymbol{p}}_j$ and half-angle $\arcsin(S_h/\|\boldsymbol{r}\|)$; we evaluate it on a 1° heading grid for a few ground speeds. The free set $\mathcal{F}_i = \complement \bigcup_j \mathrm{VO}_j$ is reported as arcs of headings. It assumes straight flight and is used as a quick picture — for the model and for the trainee (Section 9) — not as the solver's criterion, which is route-aware.

6 Solution sets

6.1 Control space and separation function

For an aircraft i, the candidate clearances form a finite control set $\mathcal{U}_i = \mathcal{U}_i^{\mathrm{spd}} \cup \mathcal{U}_i^{\mathrm{dct}} \cup \mathcal{U}_i^{\mathrm{hdg}} \cup \mathcal{U}_i^{\mathrm{lvl}}$:

- speed: from the present Mach M_0, $M_0 + \delta$ for $\delta \in \{-0.02, \ldots, -0.06, +0.01, +0.02, +0.03\}$ within $[0.68,\ \min(0.86, M_{\mathrm{MO}} - 0.01)]$; or from the indicated airspeed, ± 20 to ± 50 kt within $[210, 340]$;
- direct: the second to fourth points of the route ahead;
- heading: $\psi_i + \Delta\psi$, $\Delta\psi \in \{\pm 10, \pm 20, \pm 30, \pm 45, \pm 60, \pm 90\}$, for $\tau \in \{120, 240, 360\}$ s, then resume;
- level: ± 1000 to ± 4000 ft from the cleared level, of the aircraft's direction (semicircular rule) unless an arrival descends, under its ceiling.

Each candidate is first submitted to the simulated pilot's own check $\mathcal{P}_i$ (performance at its level and weight, weather, RVSM approval): a clearance the pilot would refuse is not a solution.

Definition 3 (Separation function)**.** *For $u \in \mathcal{U}_i$, with X_i^u the trajectory under u and X_j the predicted trajectories of the others,*

$$\Phi_i(u) = \min_{j \neq i} d^*\big(X_i^u, X_j\big). \tag{11}$$

Definition 4 (Solution set)**.** *With a margin m ($m = 1$ NM),*

$$\mathcal{S}_i = \{u \in \mathcal{U}_i \cap \mathcal{P}_i : \Phi_i(u) \geq S_h + m\}. \tag{12}$$

$\mathcal{S}_i$ is the super-level set $\{\Phi_i \geq S_h + m\}$ of Φ_i restricted to the flyable controls: the level-set view of the problem. Its boundary $\{\Phi_i = S_h + m\}$ separates, in the plane of a delaying vector's heading change and duration, the vectors that work from those that do not (Fig. 3). Because Φ_i takes the minimum over *all* other aircraft, requirement (N) holds by construction:

Proposition 3 (No new conflict)**.** *If $u \in \mathcal{S}_i$ and the other aircraft follow their predicted trajectories, aircraft i under u has no predicted conflict with any aircraft over $[0, H]$, with margin m.*

Evaluating $\Phi_i(u)$ costs $O\big((n-1)\bar{K}\big)$ by Lemma 1; with early exit as soon as one pair falls under $S_h + m$, rejected candidates are cheap.

Algorithm 1: Solution sets of a situation

Input: world w, horizon H, margin m, rollouts k

$X \leftarrow$ predicted trajectories of all active aircraft;
$\mathcal{C} \leftarrow$ conflicts of $C(X)$ involving an assumed aircraft, ordered by t^f;
foreach $(a, b) \in \mathcal{C}$ **do**
  decide the aircraft to move by (P); order = (mover, other);
  foreach *i in order* **do**
    if $t^f \le T_{\mathrm{HJ}}$*, both level, and* $V_{\mathrm{HJ}}(i, \cdot) \le \mu$ **then** drop heading and direct candidates;
    foreach $u \in \mathcal{U}_i \cap \mathcal{P}_i$ **do**
      build X_i^u; $\Phi \leftarrow \min_{j \neq i} d^*(X_i^u, X_j)$;
      if $\Phi \ge S_h + m$ **then** add u to $\mathcal{S}$ with cost $J(u)$;
  if $\mathcal{S} = \emptyset$ **then** search pairs of the best partial moves;
  sort $\mathcal{S}$ by J; confirm by rollout the first ones until k are confirmed (at most $k + 4$);
  drop refuted solutions; $\mathcal{O} \leftarrow$ optimal set;

6.2 Costs, priority and the asking rule

The controller's order (O) is encoded in a cost whose kind offsets dominate the magnitude terms:

$$J(u) = \begin{cases} 0.08\,|\delta|/0.01 & \text{speed (per 0.01 Mach or 10 kt)} \\ 1.0 + 0.05\,k & \text{direct to the } (k{+}1)\text{-th point ahead} \\ 1.2 + 0.06\,|\Delta\psi|/10 + 0.05\,\tau/120 & \text{delaying vector} \\ 3.0 + 0.15\,|\Delta L|/1000\ \text{ft} + 0.5\,\mathbb{1}_{\mathrm{ask}} & \text{level} \end{cases} + \; 5 \cdot \mathbb{1}_{\mathrm{prio}}, \tag{13}$$

with $\mathbb{1}_{\mathrm{prio}} = 1$ when i has priority. Priority (P) is decided per pair: the aircraft changing level moves if the other is level; between two level aircraft, the later to enter the sector moves (a tie within 5 s gives no priority). Moving the aircraft with priority costs 5, more than any single clearance of the other, so it is proposed only when nothing else works. The asking rule (A) adds 0.5 to a level the pilot did not request and marks it *ask first*. The *optimal* set is

$$\mathcal{O} = \{u \in \mathcal{S} : \mathrm{kind}(u) = \mathrm{kind}(u^\star),\ J(u) \le J(u^\star) + \varepsilon\}, \qquad u^\star = \arg\min_{\mathcal{S}} J,\ \varepsilon = 0.3. \tag{14}$$

Reporting a set rather than $u^\star$ alone matters for grading: two directs a few miles apart, or a vector of 20° or 30°, are equally good answers.

6.3 Pairs of clearances

When neither aircraft of the conflict has a single solution, the solver keeps, for each, the six candidates that increase most its distance to the other, and searches the 6×6 pairs (u_i, u_j) such that $\Phi_i(u_i) \ge S_h + m$ given $X_j^{u_j}$ and $\Phi_j(u_j) \ge S_h + m$ given $X_i^{u_i}$, at cost $J(u_i) + J(u_j) + 1$. If none exists the conflict is reported as needing a combination or action on third traffic.

7 Close range: Hamilton–Jacobi reachability

The straight-line and route-aware predictions treat a turn as an instantaneous change of track after a delay. Close in, the turn itself decides the outcome. For a pair, we ask: *can the moved aircraft, by turning alone at up to a standard rate, still avoid the other, which flies straight on?*

Relative dynamics. In the frame of the moved aircraft (1), with x along its track and y to its right, let (x, y) be the position of the other (2) and $\theta = \psi_2 - \psi_1$ the track difference (clockwise, as tracks are). With speeds v_1, v_2 and the turn rate u of aircraft 1, $|u| \le \omega$:

$$\dot{x} = -v_1 + v_2 \cos\theta + u\, y, \qquad \dot{y} = v_2 \sin\theta - u\, x, \qquad \dot{\theta} = -u. \tag{15}$$

The target is the loss of separation, $\mathcal{T} = \{l(\boldsymbol{\xi}) \le 0\}$ with $l = \sqrt{x^2 + y^2} - S_h$ and $\boldsymbol{\xi} = (x, y, \theta)$.

Backward reachable tube. The set of states from which a loss cannot be avoided within T whatever the turn is $\{V(\cdot, T) \le 0\}$, where V is the viscosity solution [9] of the Hamilton–Jacobi–Isaacs variational inequality

$$\frac{\partial V}{\partial \sigma} = \min\Big(0,\ \mathcal{H}\big(\boldsymbol{\xi}, \nabla V\big)\Big), \qquad V(\boldsymbol{\xi}, 0) = l(\boldsymbol{\xi}), \tag{16}$$

in the backward time σ, the $\min(0, \cdot)$ freezing the tube: once unavoidable, unavoidable. The aircraft maximises the distance, so

$$\mathcal{H}(\boldsymbol{\xi}, \boldsymbol{q}) = \max_{|u| \le \omega} \boldsymbol{q} \cdot \boldsymbol{f}(\boldsymbol{\xi}, u) = q_1(-v_1 + v_2 \cos\theta) + q_2 v_2 \sin\theta + \omega \left| q_1 y - q_2 x - q_3 \right|, \tag{17}$$

since $\boldsymbol{q} \cdot \boldsymbol{f}$ is affine in u with slope $q_1 y - q_2 x - q_3$. The optimal turn is bang-bang, $u^\star = \omega\, \mathrm{sgn}(q_1 y - q_2 x - q_3)$, and gives the direction of escape.

Numerical scheme. Equation (16) is solved on a $101 \times 101 \times 60$ grid (the formulation and scheme follow [6, 8]) over $[-45, 45]^2\,\mathrm{NM} \times [0, 2\pi)$, periodic in θ, by a first-order Lax–Friedrichs scheme with central differences $\bar{q}$ and *local* dissipation coefficients

$$\alpha_x = |-v_1 + v_2 \cos\theta| + \omega |y|, \quad \alpha_y = v_2 |\sin\theta| + \omega |x|, \quad \alpha_\theta = \omega, \tag{18}$$

$$V^{n+1} = V^n + \Delta\sigma\, \min\Big(0,\ \mathcal{H}(\boldsymbol{\xi}, \bar{\boldsymbol{q}}) + \tfrac{1}{2} \textstyle\sum_d \alpha_d \left(q_d^+ - q_d^-\right)\Big), \tag{19}$$

with the time step set by a CFL number of 0.8: $\Delta\sigma = 0.8 / \big(\bar{\alpha}_x / \Delta x + \bar{\alpha}_y / \Delta y + \alpha_\theta / \Delta\theta\big)$. A global dissipation coefficient, set by the turning term at the edge of the grid (ωL), smeared the solution over several miles; the local one does not. The horizon is $T = 300\,\mathrm{s}$, $v_1 = 450\,\mathrm{kt}$, and one table is computed per speed ratio $v_2 / v_1 \in \{0.8, 0.9, 1.0, 1.1, 1.2\}$ and per minimum S_h, once (about a minute each), and stored; a lookup is a trilinear interpolation.

Use. For a conflict with $t^f \le T$ in which both aircraft are level, the solver looks up V at the pair's relative state; if $V \le \mu$ for the aircraft to move, no turn of that aircraft separates the pair in time and its heading and direct candidates are dropped (only speed — rarely enough at that range — or a level remain). The threshold $\mu = 1.5\,\mathrm{NM}$ absorbs the optimism of the first-order scheme measured in Section 10.

8 Confirmation by simulation

The candidates are ranked on the trajectory model; the best are then confirmed on the simulator itself. The world is copied (aircraft, performance and weight, wind, weather, entries, pilots), the clearance is entered as a controller would — refused if the simulated pilot would refuse it — and the copy is advanced by one-second steps over H, a delaying vector being resumed after τ. Every two seconds, the distance of each moved aircraft to every aircraft vertically too close to it is recorded; the solution is *confirmed* when the minimum stays at or above S_h, and refuted otherwise. The rollout costs 0.7 s per solution for a ten-minute horizon; the solver confirms the cheapest solutions until k ($k = 2$ by default) are confirmed, trying at most $k + 4$. Refuted solutions are removed from the set.

9 Integration in the simulator

Solo radar. A *Solver* panel lists, for each predicted conflict involving one of the trainee's aircraft, the conflict (d^* and time to first infringement), the priority reason, and the solutions in cost order with their kind, the predicted or confirmed distance, and *ask first* or *priority* marks. *Apply* enters the clearance in the label exactly as the trainee would; a delaying vector leaves a resume reminder with a countdown. The panel is refreshed every five simulated seconds or when an entry changes.

AI lab. The solver is used in three ways. (i) As a controller of its own: a built-in agent assumes and transfers aircraft, answers requests, and applies the optimal solution of each conflict, giving a reference score for each scenario. (ii) As hints in the observation given to a language model: each conflict line ends with the solver's best and other options. (iii) As a grader: each decision of any agent on a conflict is classified as *optimal* (in $\mathcal{O}$), *feasible* (in $\mathcal{S} \setminus \mathcal{O}$), *other*, or *left alone*, and the labels of the experience dataset carry a reward of $+2$, $+1$ and -1 for the first three. The model can also query the solver (`solve`) and the velocity obstacles (`headings`).

10 Results

All experiments use the simulator's own scenarios of the Agadir airspace (Moroccan upper airspace, four sectors) and its own code; the figures are drawn from simulated situations.

10.1 A worked example

Figure 1 shows the West North sector of scenario *p02* at T+300 s, with 15 aircraft on the picture and $S_h = 7$ NM. Two aircraft at FL350, EWG3KP and IBE3247, are on routes merging at a common point; the solver predicts a first infringement in 216 s and, on a longer horizon, a closest approach of 0.0 NM at 830 s: without action they would be in trail at the same point and level. They entered the sector together, so neither has priority.

Velocity obstacles. Figure 2 gives the headings of EWG3KP that bring it within S_h of another aircraft flying straight on (at its present ground speed): IBE3247 116°, NOZ4381P 37°, BAW9522 20°. Its present track 230 lies inside the obstacle of IBE3247; the free arcs are 069–221, 338–031.

Solution set. Figure 3 shows Φ over the plane of a delaying vector (heading change and duration, then resume), sampled on 555 controls (0.6 ms each), and over speed alone. Speed cannot solve this conflict: over Mach 0.71–0.82 the closest approach to anyone stays under 4.4 NM. The super-level set $\{\Phi \geq S_h + m\}$ is the region of left turns of 20° and more; right turns bring the aircraft towards other traffic. The solver returns 12 solutions; the optimal set is *direct TOVRA* (confirmed 9.4 NM) and *direct VEDOD* (confirmed 10.1 NM), both shortening the route (direct before vector in the order of preference).

The whole solution space. Figure 4 draws the control space of both aircraft of the pair, one family of clearances per column, on a common scale of Φ. Of the 99 candidates the pilots would fly, 42 leave everyone separated by the margin: speed 0 of 12, direct 2 of 6, delaying vector 32 of 72, level 8 of 9. The families behave very differently. Speed alone never reaches the margin. Only EWG3KP's shortcuts work. Among the vectors, EWG3KP must turn left by 20° or more, while IBE3247 has two separate regions — a right turn, or a left turn of 60° or more — so the solution set of one aircraft need not be connected. Nearly every level works, but each one the pilot did not request must be asked first. Figure 5 sets every candidate by its cost against Φ:

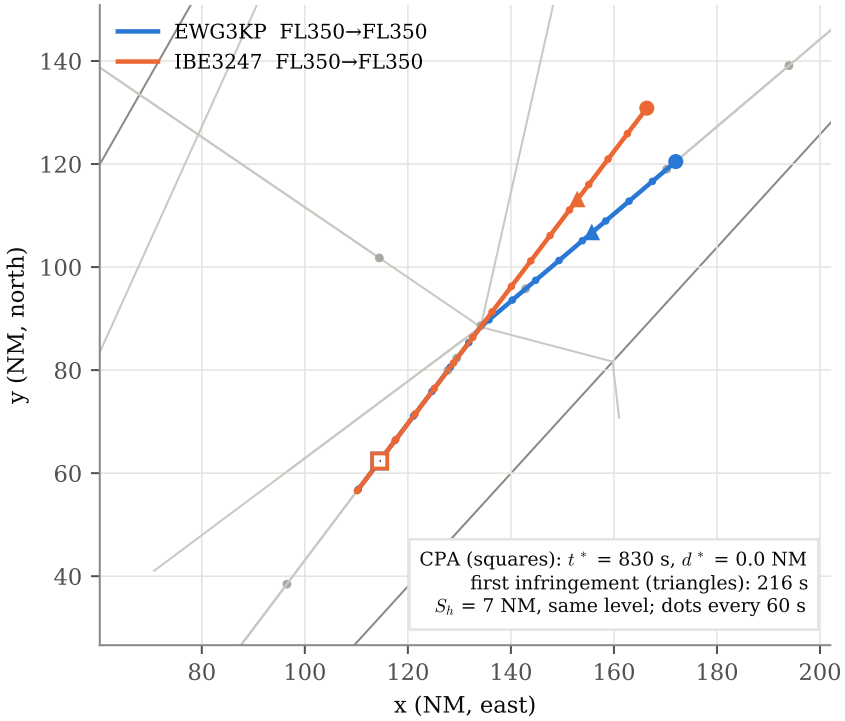


Figure 1: Predicted piecewise-linear trajectories of a situation (grey), the conflicting pair in colour. Squares: the closest point of approach; triangles: the first infringement of the minima, both exact (Lemma 1).

the solution set is the part above the line $\Phi = S_h + m$, and the optimal set (circled) its leftmost points — the cheapest kind of clearance that works, not the one that separates most.

Two clearances at once. When no single clearance works, the solver searches pairs (Section 6). Figure 7 shows what that search is over, in a busier situation: scenario s10 at T+360 s, where AFR11 and RAM602, both at FL370 with $S_h = 5$ NM, would pass 0.4 NM apart, in busy traffic: the vectors of AFR11 also bring it near RYR3, TAP7 and SWR40. Both are given a 4-minute delaying vector, each heading change on one axis, and the surface is Φ, the closest either then comes to anyone ($37 \times 37 = 1369$ combinations, 0.07 ms each). The single-clearance solutions lie on the two axes, where one aircraft keeps its route; here neither axis reaches the plane $\Phi = S_h + m$. Separating the pair alone is easy: 72% of the plane does it (dotted). Leaving everyone separated is not: the third aircraft — mainly RYR3, TAP7 and SWR40 — cut that down to 19%, in two regions. In both, AFR11 turns left; RAM602 then turns right by about 40° or more, or left by about 30° or more — far enough that neither meets the others. This is the reason the separation function takes the minimum over everyone: the dotted contour, the answer of a pairwise method, is mostly made of clearances that create a new conflict.

10.2 Exact CPA against sampling

Over 123 situations taken every two minutes in ten scenarios (aircraft not controlled), the 5793 pairs coming within 60 NM of each other were evaluated exactly and by sampling the same trajectories every Δ seconds (Table 1, Fig. 8). The error grows with Δ as (8) predicts: up to 0.70 NM at 20 s, the step of the radar's medium-term detection, and 4.02 NM at 60 s. In these situations no vertical window was lost and no conflict missed at any step: the main effect of sampling is the distance it reports, which decides whether a marginal conflict is shown and whether a clearance clears it by the margin. The windows a sampling can lose are short ones —

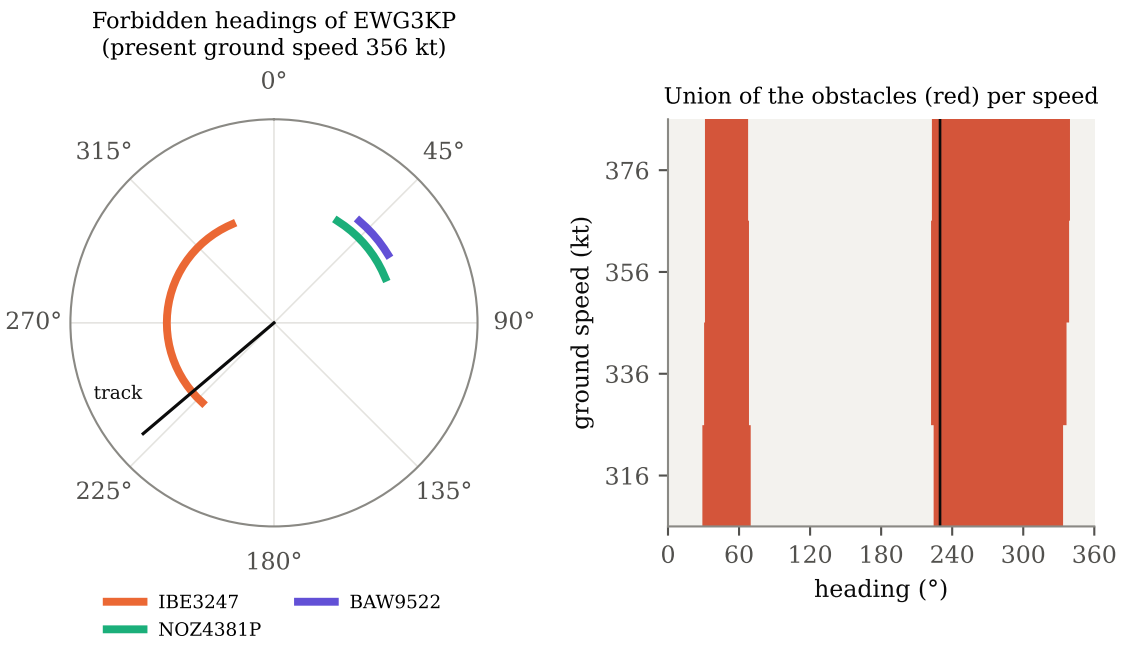


Figure 2: Velocity obstacles of the aircraft to move. Left: forbidden headings per other aircraft (arcs) and the present track. Right: their union over heading and ground speed.

Table 1: Sampled minimum minus exact minimum distance (NM) while vertically too close, over the pairs within 60 NM. *Window lost*: the sampling never sees the pair vertically too close although it is; *conflicts missed*: exact conflicts (of 66) the sampled distance does not show.

Δ (s)	median	95th pct.	max	window lost	conflicts missed
1	0.00	0.00	0.04	0	0
5	0.00	0.01	0.21	0	0
10	0.00	0.03	0.35	0	0
20	0.00	0.06	0.70	0	0
30	0.00	0.29	1.95	0	0
60	0.00	1.02	4.02	0	0

a climb or descent crossing another level — which these uncontrolled situations contain few of.

10.3 Reachability

Figure 9 shows slices of the zero level set of V for $S_h = 5$ NM and equal speeds. Without turning, any aircraft in the band ahead whose straight path crosses the disk is a conflict; with turns at standard rate, the set from which the conflict cannot be avoided within five minutes shrinks to a region a few miles ahead of the moved aircraft, longest head-on.

To validate the table and the frame used to look it up, 2422 relative states within 25 NM were drawn at random and the encounter simulated in the plane: the moved aircraft turns left or right at 3°/s for 0 to 300 s and then flies straight, the other flies straight, and the best of these strategies gives a lower bound of the true value. The sign of V agrees with the simulation in 99.9% of the states (Fig. 10). Near the boundary the table is slightly optimistic (mean 0.28 NM, median 0.05 NM, part of which is the simulation's restricted strategies). With the threshold $\mu = 1.5$ NM, no state the simulation finds unavoidable (18) is declared avoidable by turning, at the price of 46 avoidable states declared unavoidable — the safe side.

Remark 2. *The validation found an error in an earlier version: the relative state was expressed*

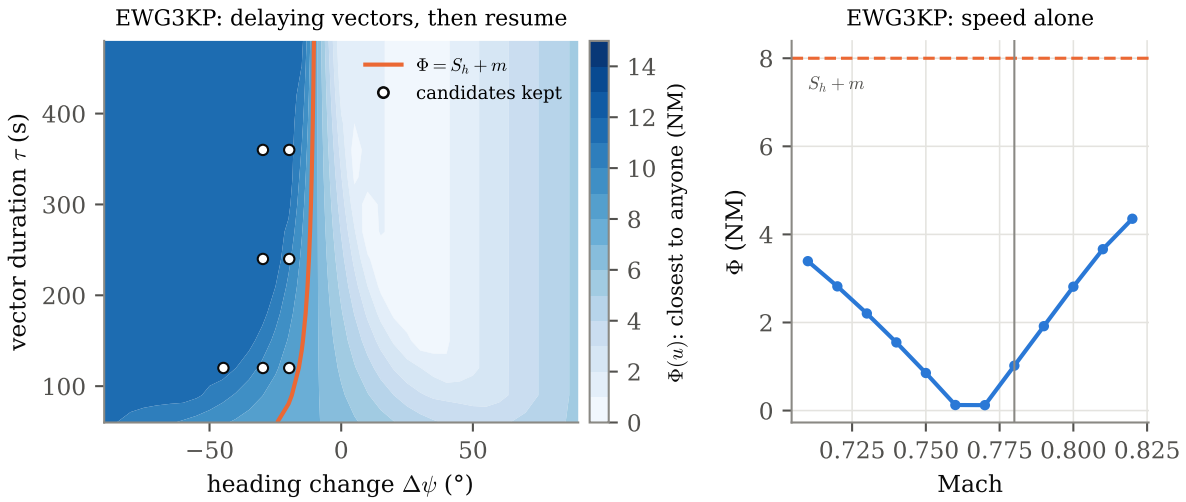


Figure 3: The separation function Φ of the aircraft to move. Left: over delaying vectors; the solution set is the region beyond the orange level line $\Phi = S_h + m$; circles: the vectors of the solver's control grid kept as solutions. Right: over speed alone (all below $S_h + m$).

with y to the left of the track while the tables' dynamics, with clockwise track angles, hold for y to the right — the lookup was of the mirror-image encounter. On the 333 states where the two frames differ by more than 1.5 NM*, the mean error is 0.60* NM *in the right frame and 2.47* NM *in the mirrored one.*

10.4 The solver on frozen situations

On the same situations (1–30 aircraft each), predicting all trajectories and computing the conflict matrix took a median of 3 ms (at most 28 ms); the cost grows with the number of pairs (Fig. 11, left). As no one controls these situations, the same encounter appears in several of them until it becomes a loss: the solver met 10 distinct encounters in 5 scenarios, 64 times in all: 39 times ahead of the encounter, and 25 times already in a loss of separation (no clearance can make $\Phi \geq S_h + m$ there: the distance is already below the minimum, and the observation instead gives the levels that separate soonest). It found solutions for 37 of the 39 future conflicts (95%); the best solution's kind was: heading 28, direct 8, speed 1. The 2 others were close (median time to infringement 40 s), where only a combination or action on the other traffic remains. A complete solve, including the rollouts, took a median of 1.1 s per situation with conflicts (at most 12.4 s) on a shared two-core machine. Each rollout of a ten-minute horizon costs about 0.7 s. For the 71 solutions confirmed by rollout, Fig. 11 (right) compares the predicted margin over the minimum with the margin in the simulation. The simulation is mostly the more favourable — the prediction keeps every aircraft on its route while the pilots of the copy also resume, turn and level off as they fly — and in 24 cases it is more than half a mile less favourable, at worst by 3.8 NM — more than the margin m covers, which is why the best solutions are confirmed before they are proposed.

10.5 Closed loop: the solver as a controller

Finally the solver controlled 12 scenarios for 20 simulated minutes each through the AI lab, against the built-in baseline controller, which assumes, transfers and answers requests the same way but resolves conflicts with its own rules (speed, then a probed heading, then a level). Both were scored by the lab's evaluation (100 minus penalties: 25 per loss of separation, 10 per infringement of an active area, track miles, late assumes and transfers…). Table 2 gives the

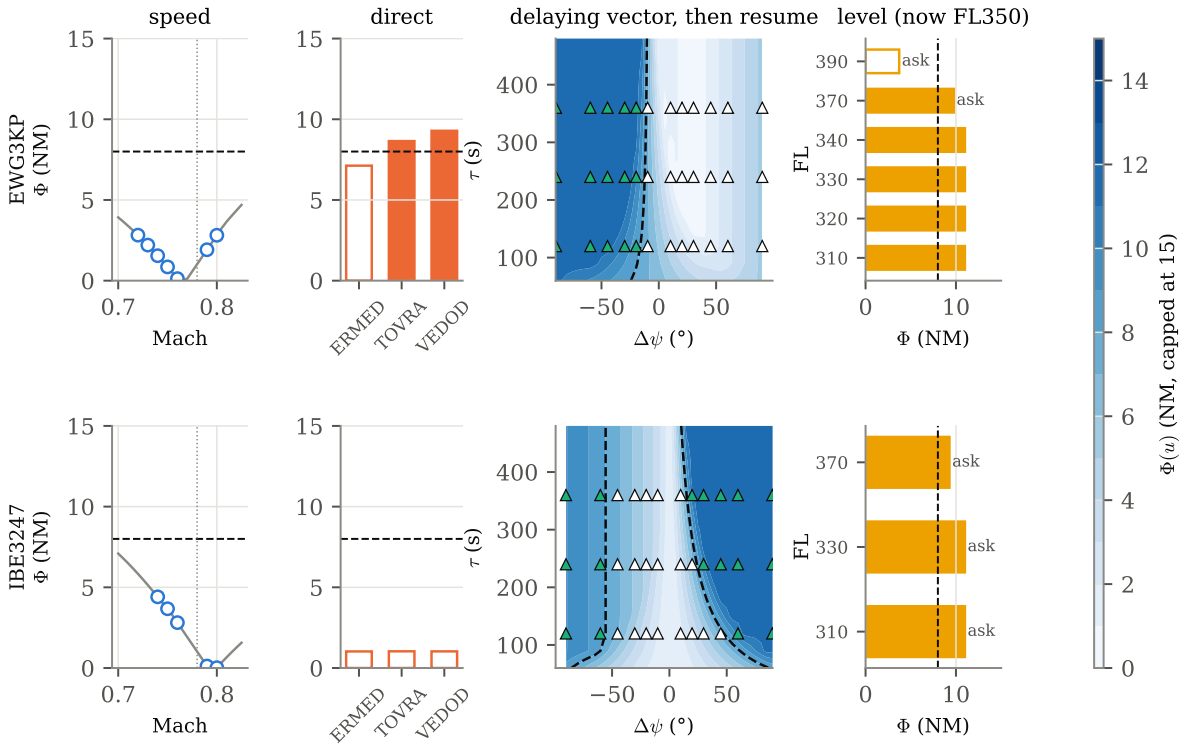


Figure 4: The solution space of the worked example, for each aircraft of the pair (rows) and each family of clearances (columns): $\Phi(u)$, the closest the aircraft then comes to anyone, on a common scale (capped at 15 NM). Dashed: $\Phi = S_h + m$. Filled markers and bars: candidates in the solution set; open: candidates that leave a conflict. Speed: the continuous curve over Mach and the candidates of the grid; direct: the points ahead on the route; vector: Φ over heading change and duration with the grid's candidates; level: the levels of the aircraft's direction it can fly (*ask*: not requested, to be asked first).

results. Mean score: solver 97.1 (0 losses in all), baseline 92.8 (2 losses). Graded against the solver's solution sets, the baseline's conflict decisions were: left alone 62, other 17, optimal 14, feasible 3 (the baseline acts on conflicts under eight minutes only, hence the many conflicts left alone at a given decision).

Neither controller handles the activation of restricted areas (scenario s10) or the non-RVSM aircraft of p05, which cost both the same penalties; the difference between them is in the losses of separation. The closed loop was also how two weaknesses of an earlier version of the trajectory model were found. In p06 the solver controller approved a pilot's request for descent that the radar's probe, with a nominal 2000 ft/min, found clear; the aircraft, high on its profile, descended at idle with speedbrakes, at almost twice that, through the level of an aircraft 5 NM away, and was mid-turn after a point. The prediction now flies turns in progress (Section 3) and checks level changes at the bounds of the rates (Proposition 1); requests are approved only when both the probe and the solver's check find them clear.

11 Discussion and limitations

Greedy per conflict. Each conflict is solved against the predicted trajectories of everyone else, one at a time. When two conflicts share an aircraft, applying the first solution changes the second; the solver is then simply run again on the new picture (every five seconds in the panel, at every decision in the AI lab). A joint formulation — a mixed-integer programme over the controls of all aircraft with the pairwise CPA as constraints — would find solutions this misses, at a cost that grows quickly with the number of aircraft.

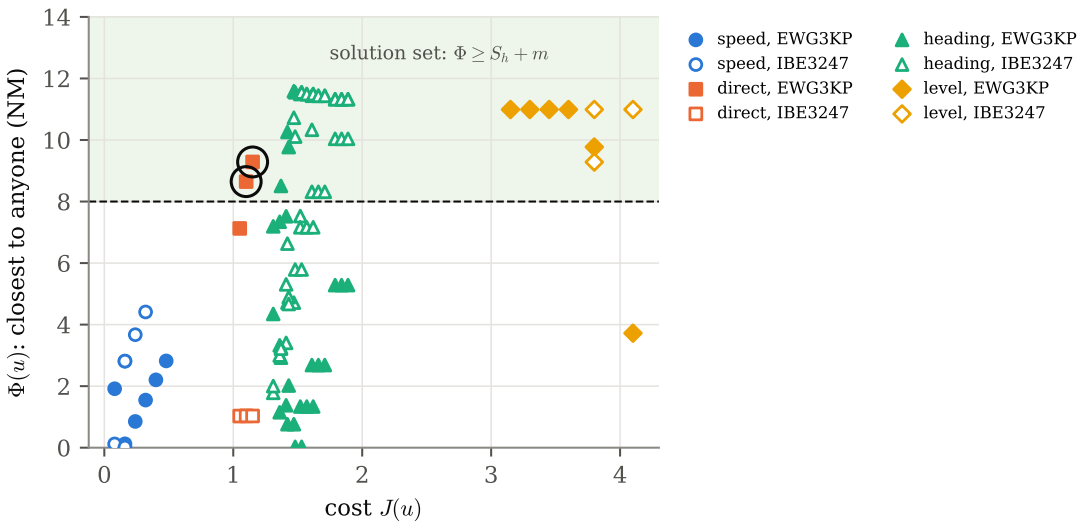


Figure 5: Every candidate of the worked example by its cost $J(u)$ and its separation $\Phi(u)$ (filled: EWG3KP; open: IBE3247). Shaded: the solution set. Circled: the optimal set.

Discrete control space. The super-level set is sampled on a grid of a few dozen controls per aircraft. The boundary $\{\Phi = S_h + m\}$ could be located more finely, for instance by bisection along a heading change or a speed, to propose the smallest sufficient deviation rather than the smallest of the grid.

Model errors. Trajectories assume the present wind along the path, constant speeds between changes and nominal vertical rates; other controllers' intentions are unknown. The margin m, the confirmation by rollout and the re-solve at every update address these, but the solution set is only as good as the prediction of the others. The Hamilton–Jacobi table is computed for a single nominal speed of the moved aircraft and five speed ratios, without wind, and is first-order accurate.

What the solver does not see. Restricted areas that become active, weather cells and the non-RVSM status of an aircraft enter the solution set only through the pilot's own check of a clearance, not as constraints of their own; the solver does not plan a route around an area or a cell. Scenario s10 shows the cost: both controllers incurred three area infringements.

Costs. The weights of J encode the unit's order of preference and priority rule; they are hand-set. Recorded controller decisions — from the solo radar and the exercises — could be used to fit them, the solution set providing the alternatives against which each human choice is compared.

12 Conclusion

Exact CPA on piecewise-linear trajectories turns conflict detection into a matrix computation and makes the quality of any candidate clearance a single number, $\Phi(u)$, the closest it brings the aircraft to anyone. The solution set is the region of the control space where that number clears the minimum with a margin, ranked by the controller's conventions, filtered at close range by reachability, and confirmed by simulation. It serves the trainee as advice, a language model as guidance and grading, and the simulator as a controller of reference.

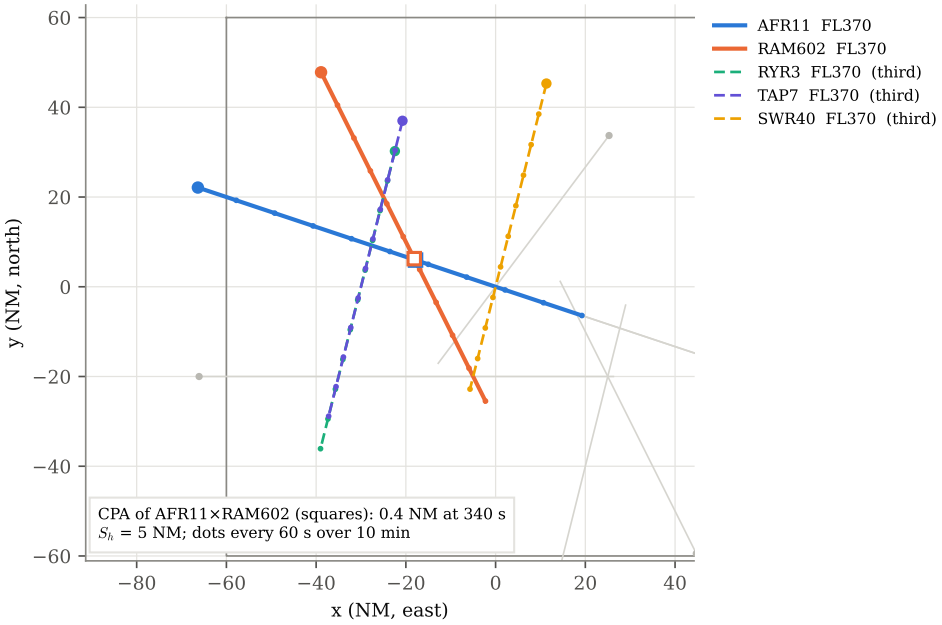


Figure 6: The traffic of the second example (scenario s10, T+360 s): predicted trajectories over ten minutes, all at FL370. Solid: the conflicting pair; dashed: the three aircraft that constrain their vectors (RYR3 follows TAP7 on the same route); grey: the rest.

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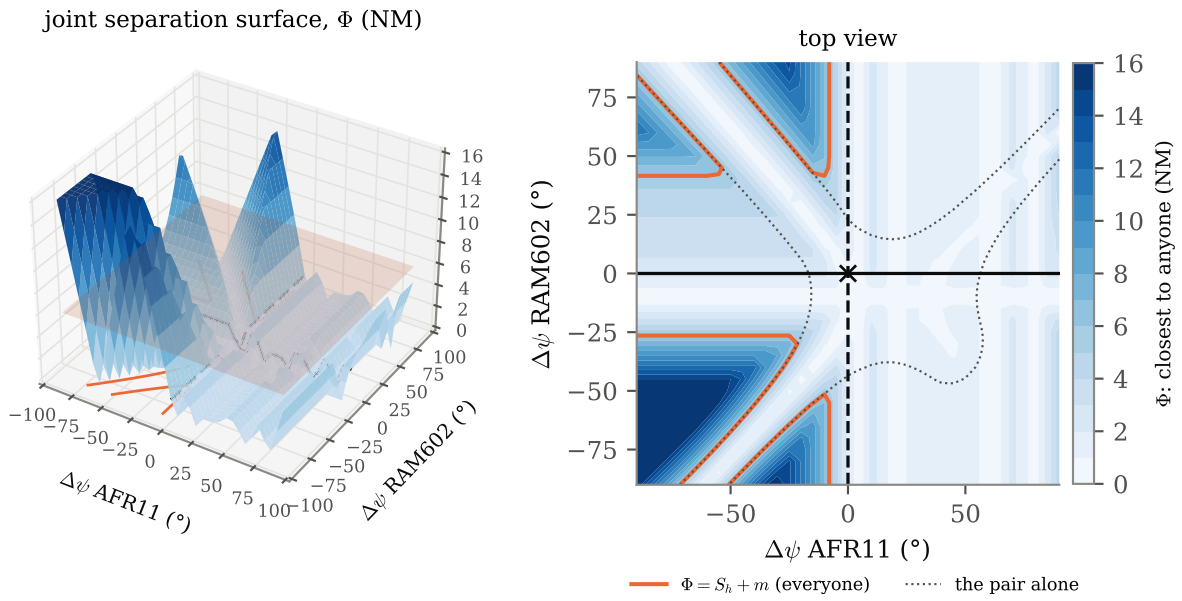


Figure 7: Joint heading space of a conflicting pair (scenario s10): both aircraft vectored for 4 minutes, then resuming. Left: the surface of Φ over the two heading changes and the plane $\Phi = S_h + m$. Right: top view; orange: the solution set's boundary, dotted: the same boundary for the pair alone, ignoring everyone else; solid and dashed axes: one aircraft keeps its route (single clearances); cross: no action.

A Parameters

Symbol	Value	Meaning
H	600 s	prediction horizon of the solver
m	1.0 NM	margin on S_h for a solution
δ_s	40 s	time to achieve a speed change
δ_p	6 s	pilot delay
ω	3°/s	standard-rate turn
τ	120, 240, 360 s	durations of a delaying vector
ε	0.3	cost tolerance of the optimal set
T_{HJ}	300 s	horizon of the reachability tables
μ	1.5 NM	threshold on V ("a turn is enough")
grid	$101 \times 101 \times 60$	$[-45, 45]^2$ NM $\times$ $[0, 2\pi)$
k	2	solutions confirmed by rollout per conflict

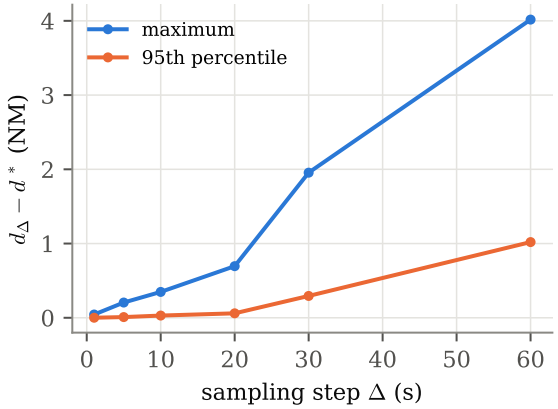


Figure 8: Error of a sampled minimum distance against the exact CPA.

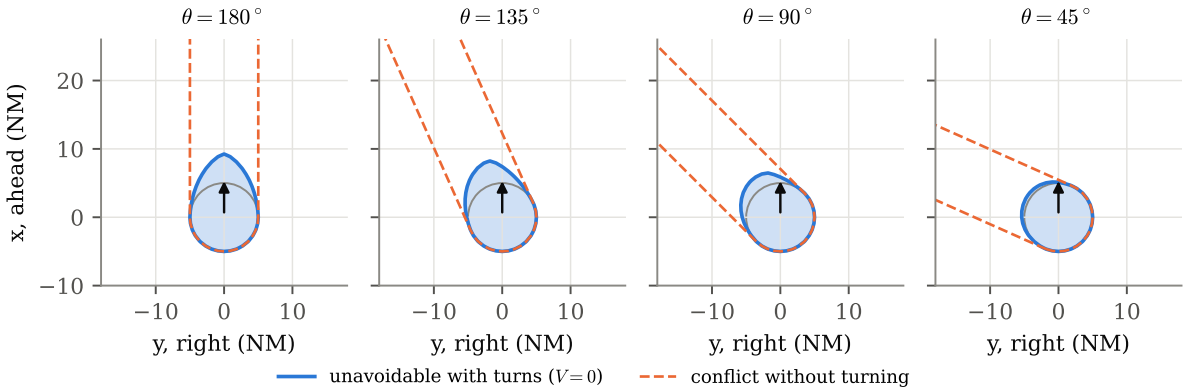


Figure 9: Slices θ = const of the reachability tables ($S_h = 5\,\text{NM}$, $v_2 = v_1 = 450\,\text{kt}$, $T = 300\,\text{s}$). The moved aircraft is at the origin heading up (arrow); the other is at (y, x) with track difference θ. Blue: the states from which no turn avoids the loss; dashed: the conflicts of straight flight; grey: the target disk.

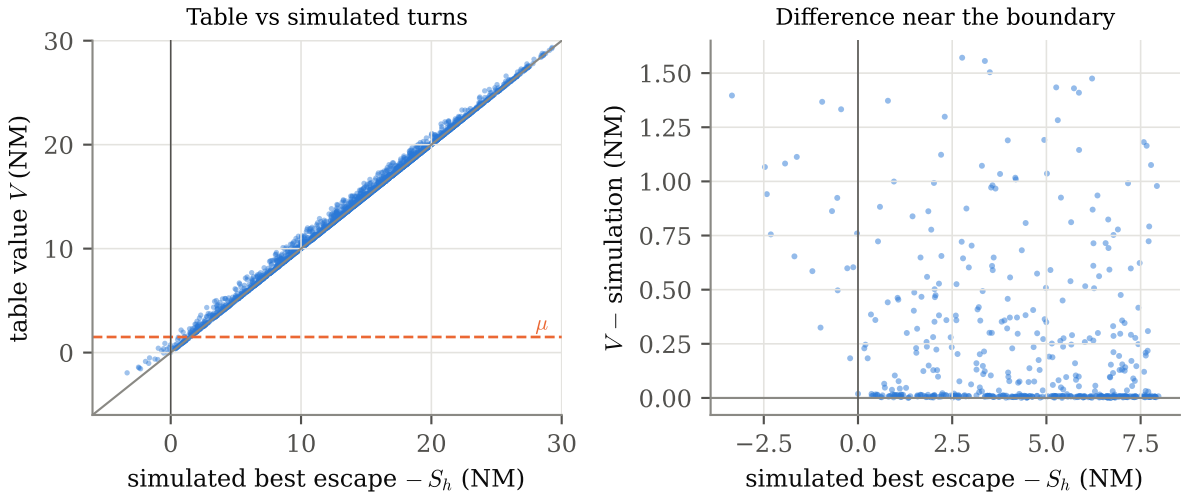


Figure 10: Validation of the reachability table against simulated turns: the table value against the best simulated escape (left), and their difference near the boundary (right).

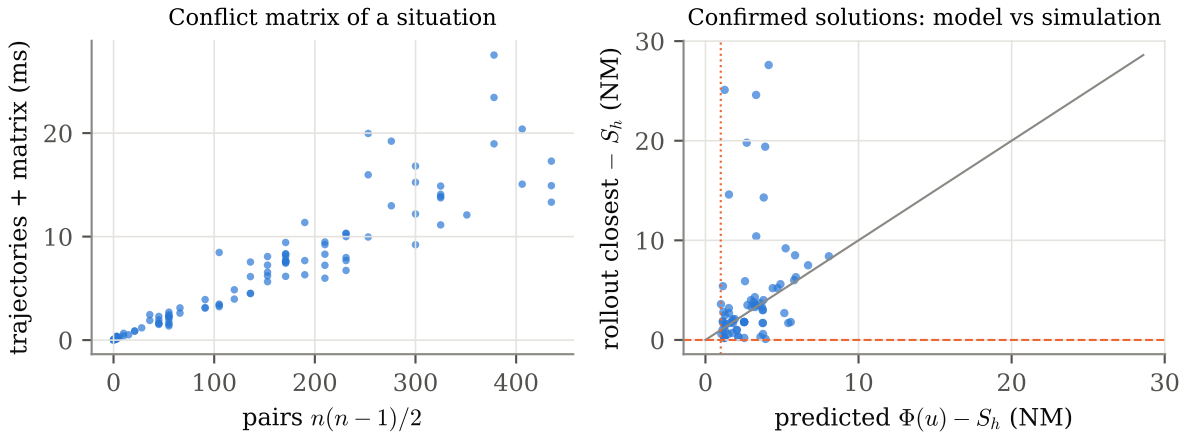


Figure 11: Left: cost of the trajectory prediction and conflict matrix of a situation against the number of pairs. Right: for the solutions confirmed by rollout, the margin over S_h predicted against the margin in the simulation (dotted: the margin m required of a solution; dashed: the minimum).

Table 2: Twenty-minute runs. Score (0–100), losses of separation, clearances given (speed / heading or direct / level) and the smallest separation in a loss (NM).

	Baseline				Solver			
Scenario	score	loss	spd/lat/lvl	min	score	loss	spd/lat/lvl	min
c01_cta_moderate_wx	100.0	0	0/1/0	–	100.0	0	0/0/0	–
p01_ws_two_merges	74.4	1	0/4/0	6.5	100.0	0	0/3/0	–
p02_wn_merges_and_climbs	100.0	0	0/3/0	–	100.0	0	9/1/0	–
p05_ws_storms_arrivals	95.0	0	0/0/0	–	95.0	0	0/0/0	–
p06_wn_capacity_sequencing	74.0	1	0/8/0	6.6	100.0	0	0/2/0	–
s01_basic_crossing	100.0	0	0/1/0	–	100.0	0	3/1/0	–
s02_climb_request_into_crossing	100.0	0	0/0/0	–	100.0	0	0/0/0	–
s03_descent_through_opposite	100.0	0	0/0/0	–	100.0	0	0/0/0	–
s05_slowdown_on_converging_exit	100.0	0	1/1/0	–	100.0	0	1/0/0	–
s06_speed_up_into_crossing	100.0	0	3/1/0	–	100.0	0	1/1/0	–
s08_simultaneous_level_requests	100.0	0	0/0/0	–	100.0	0	0/0/0	–
s10_busy_mixed	70.0	0	7/7/0	–	70.0	0	1/7/0	–
Mean	92.8	2			97.1	0		